DEEP LEARNING FOR SEQUENTIAL DATA & NLP · FINAL PROJECT

NLP core of a contract-review assistant

Classify clauses, retrieve precedents, and document where the system can — and cannot — be trusted. Data: LEDGAR (LexGLUE).

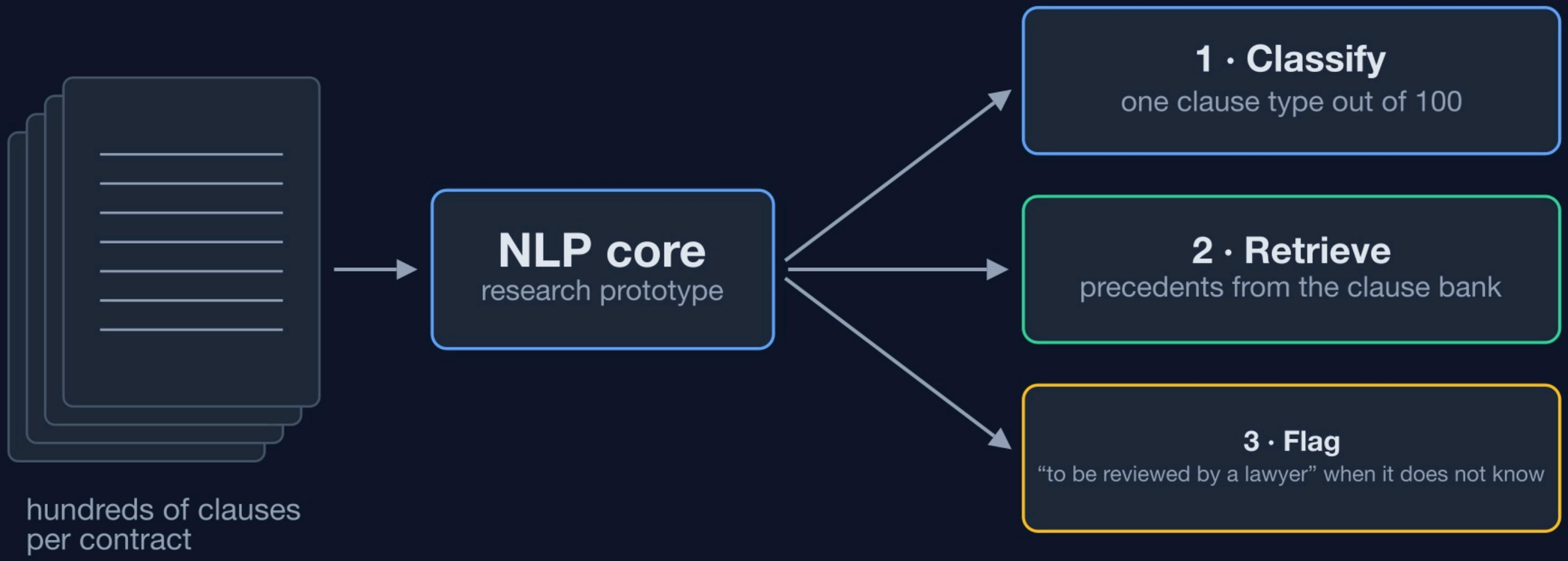
Research prototype · English clauses (SEC) · not a substitute for a lawyer

Louis · problem, data, conclusion

Charles · classification

Alessandro · retrieval, audit, reproducibility

Due diligence: every clause is read, typed and compared



We do not automate the lawyer: we triage, retrieve and flag.

LEDGAR: 80,000 clauses from contracts filed with the SEC



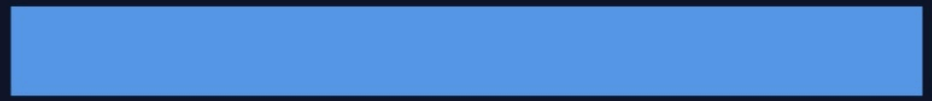
100 clause types | **1 label per clause** | **legal English** | **US law (SEC)**

Chronological split: the test already measures some drift (Jensen-Shannon train/test 0.078)

Three limitations measured before modelling

1 · Imbalance (train)

Governing Laws · 3,167



Books · 23

ratio 138

→ we track macro-F1 (each type weighs the same)

2 · Scope

English · US law

non-US jurisdiction: 253 clauses (train)

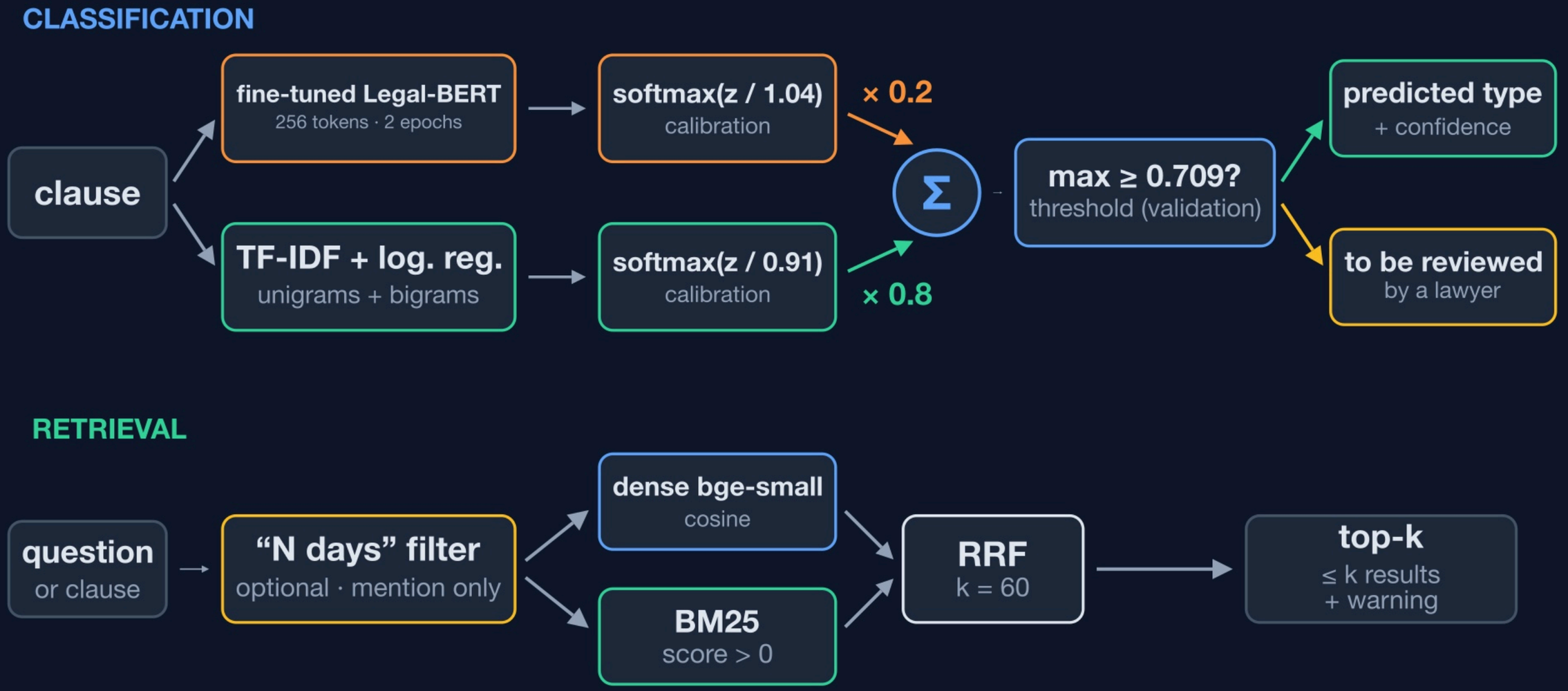
3 · Types missing from the 100 labels

- × GDPR / data protection
- × force majeure
- × non-compete
- × limitation of liability
- × exclusivity · audit · non-solicitation

the system cannot recognise them

Also: 282 test clauses identical to a training clause → scores published with and without them

Architecture: two pipelines, one shared interface



Temperatures, weight and thresholds: chosen on validation only. The test is only used for measurement.

Baseline: TF-IDF + logistic regression

What it is

- TF-IDF vectors over unigrams and bigrams
- Logistic regression over the 100 types
- ≈ 4 min of training on CPU

Why it is hard to beat

On LEDGAR, a clause's type often hinges on a few keywords: “governing law”, “counterparts”, “severability”. A lexical model captures them very well.

87.70
test micro-F1 [87.02; 88.37]

82.19
test macro-F1 [80.52; 83.00]

Published reference (LexGLUE, Chalkidis et al. 2022, Table 3): TF-IDF + SVM **87.0 / 81.4**.

Indicative comparison: same test, but a different protocol (1-3-gram SVM with grid search).

Fine-tuning Legal-BERT

CHOICE	VALUE	REASON
Model	legal-bert-base-uncased	legal English; CamemBERT is French
Length	256 tokens	88% of clauses kept whole; 512 saturated 16 GB
Budget	2 epochs · 1 seed (42)	Apple M3 laptop, 146 min
Optim.	AdamW $3 \cdot 10^{-5}$, warm-up 6%, bf16	standard BERT recipe
Loop	hand-written PyTorch	length-bucketed batches, resumable checkpoints

Validation during training

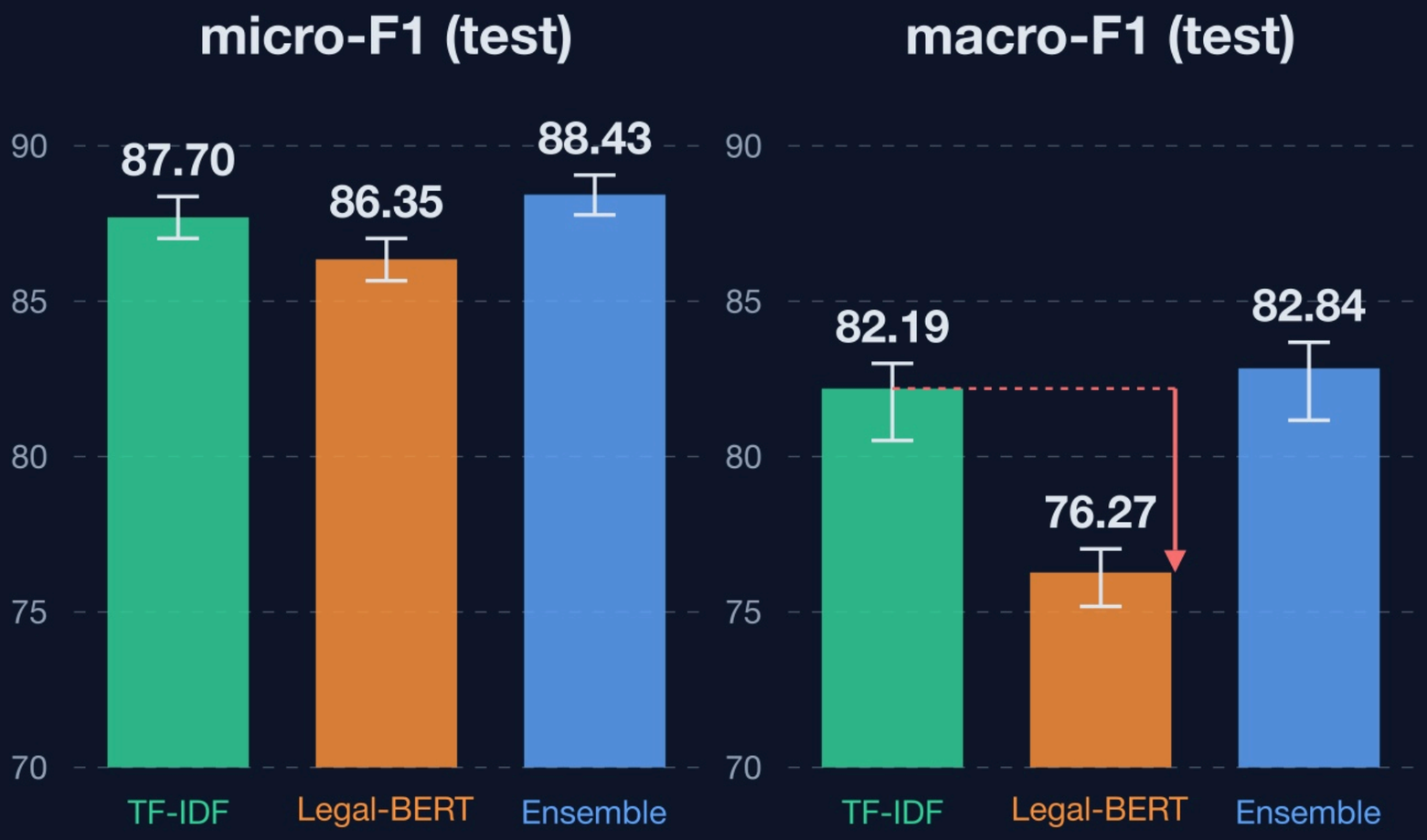
macro-F1: **74.0** (epoch 1) → **77.6** (epoch 2)

It was still rising. Values from the training trace (definition used at the time: present labels only).

Provenance

A single process, step 0 → 7,500, with no resume (proven by the log). Resume tested on CPU only.

LEDGAR test (10,000 clauses): three variants



Legal-BERT – TF-IDF
macro –5.92 pt [–6.88; –4.50]
micro –1.35 pt [–1.93; –0.77]
BERT alone drops off on rare types

Ensemble – TF-IDF
micro +0.73 pt [+0.51; +0.98]
macro +0.65 pt [+0.31; +1.00]
real but modest gain

Axis truncated at 70 · whiskers = 95% CI, per-clause bootstrap (probably somewhat too narrow)

Calibrated ensemble, then a threshold decision

$$p = 0.2 \cdot \mathrm{softmax}(z_{\mathrm{BERT}} / 1.04) + 0.8 \cdot \mathrm{softmax}(z_{\mathrm{TF\text{-}IDF}} / 0.91)$$

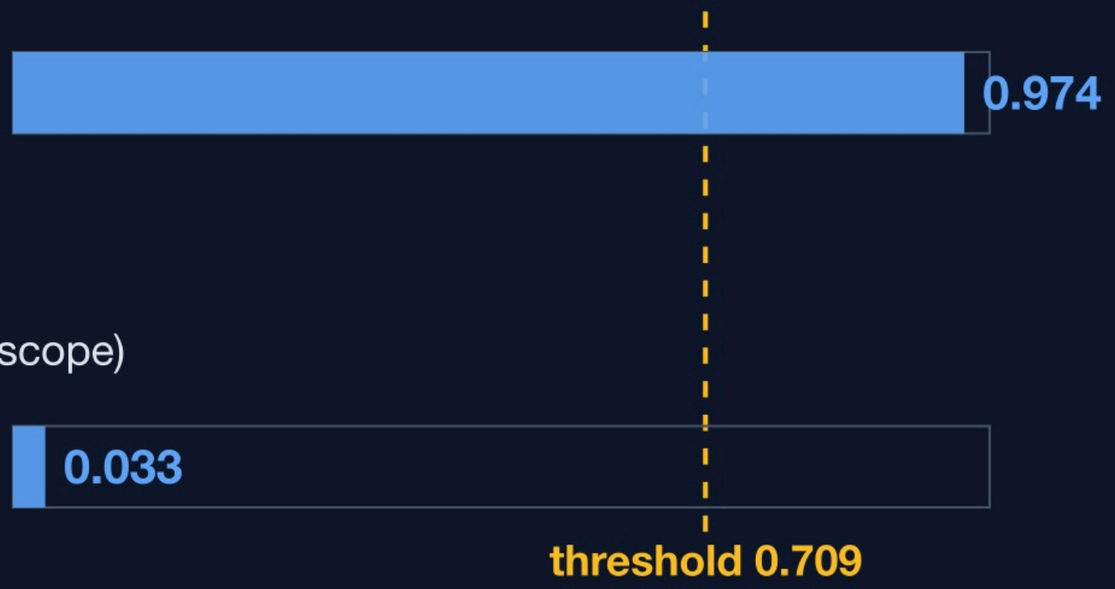
weight: grid 0 → 1 (step 0.05), best validation macro-F1 · temperatures learned on validation

"This Agreement shall be governed by … the laws of the State of New York …"

Governing Laws

≥ threshold → type returned

2nd candidate: Applicable Laws 0.023 (near-synonym)



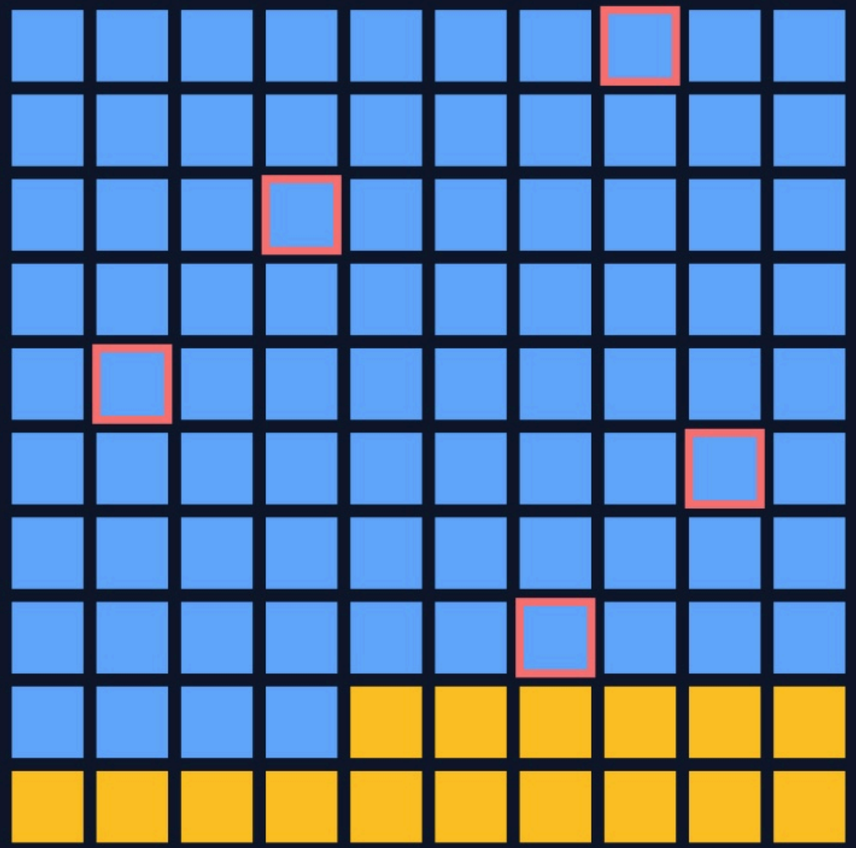
"Le présent contrat est régi par le droit français." (out of scope)

Amendments

< threshold → ABSTAIN: to a lawyer

Real recorded outputs (demo_outputs) · confidence = the TYPE is correct, not legal validity

Abstention: out of 100 test clauses (ensemble)



1 square ≈ 1% of clauses (rounded)

threshold 0.709 set on validation
(target: 95% correct classifications)

handled: 84.2% of clauses

of which 94.5% correctly classified

sent to the lawyer: 15.8%

errors nonetheless: ≈ 4.6%

(= 84.2% × 5.5%)

95% is a validation target, not a guarantee.

Test ECE 0.017 · validation coverage 84.1% → test 84.2%

Hybrid retrieval: dense + BM25, RRF fusion



bank: 59,080 training clauses, exact duplicates removed

Clause → similar clauses · 981 test queries · relevant = same LEDGAR label

	P@1	P@10	MRR@10
Dense	76.6%	64.8%	0.814
BM25	77.0%	64.6%	0.817
Hybrid	**77.4%**	**66.4%**	**0.824**

Hybrid beats each engine alone, narrowly · 40 hand-written questions already seen: non-regression

Successful example: wording close to a clause

```
$ nlp-final-project search "either party may terminate on thirty days written notice" -k 5

(period filter: clauses mentioning any of [30] day(s); 824 eligible. Mentioning a duration does not mean it is the requested notice period: the role of the duration is not checked)

[1] Terminations … either party may … terminate this Agreement at any time upon thirty (30) days' written notice to the other party …
[2] Terminations The Term of this Agreement may be terminated by either party upon thirty (30) days' prior written notice.
[3] Terminations Either party may terminate the employment relationship for any reason at any time upon giving the other party thirty (30) days prior written notice …
[4] Terminations … the breaching party shall be given not less than thirty (30) days' prior written notice of such material breach …
[5] Terminations Either the Executive or the Company may terminate … upon not less than thirty (30) days prior written notice.
```

5 / 5 results of type *Terminations*, each with a thirty-day notice.

Real recorded output (presentation/demo_outputs/3_search_success.txt), abridged excerpts. One favourable case does not validate retrieval.

Controlled failure: the “30-day notice” question

Wording close to a clause

“either party may terminate on thirty days written notice”

1. Terminations	2. Terminations	3. Terminations	4. Terminations	5. Terminations

Lawyer's wording, same intent

“show me termination clauses with a 30-day notice period”

1. Notices	2. Notices	3. Terms	4. Terminations	5. Disability

P@1 = 0 · only 1 relevant result in the top 10

Controlled failure: the “30-day notice” question

1. Notices | 2. Notices | 3. Terms | 4. Terminations | 5. Disability

Result no. 1 (label Notices) — excerpt of the returned clause

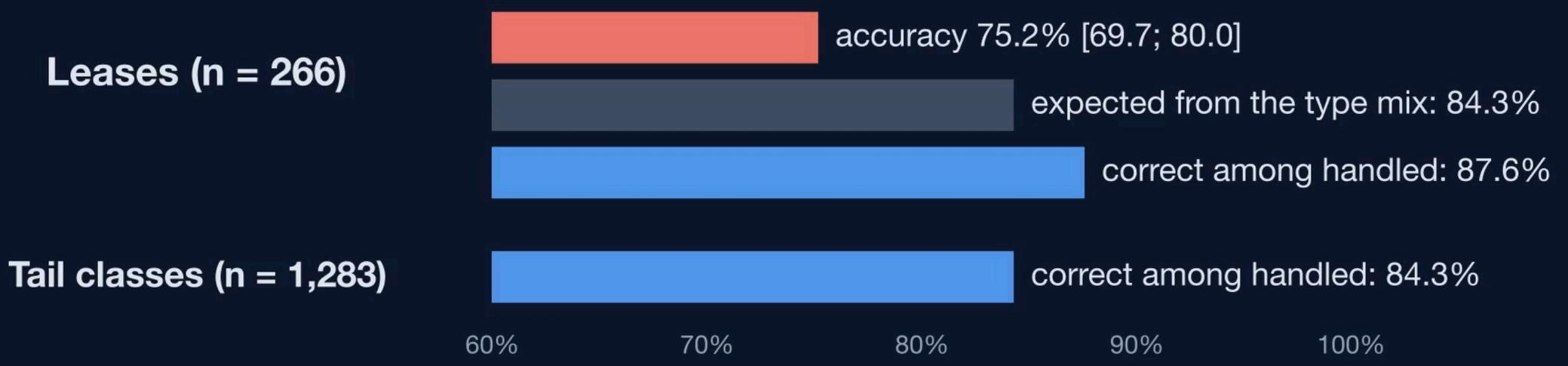
In the event Participant believes that a “Good Reason” event has been triggered, Participant must give the Company written notice within 30 days of the first occurrence of such triggering event and a proposed termination date which shall be not sooner than 60 days nor later than 90 days after …

30 days = deadline to give notice of a “Good Reason”
≠ termination notice period

The filter checks that a duration is MENTIONED, not its role (nor business / calendar days)

Audit: where the system is fragile (ensemble)

1 · Groups (keyword proxies)

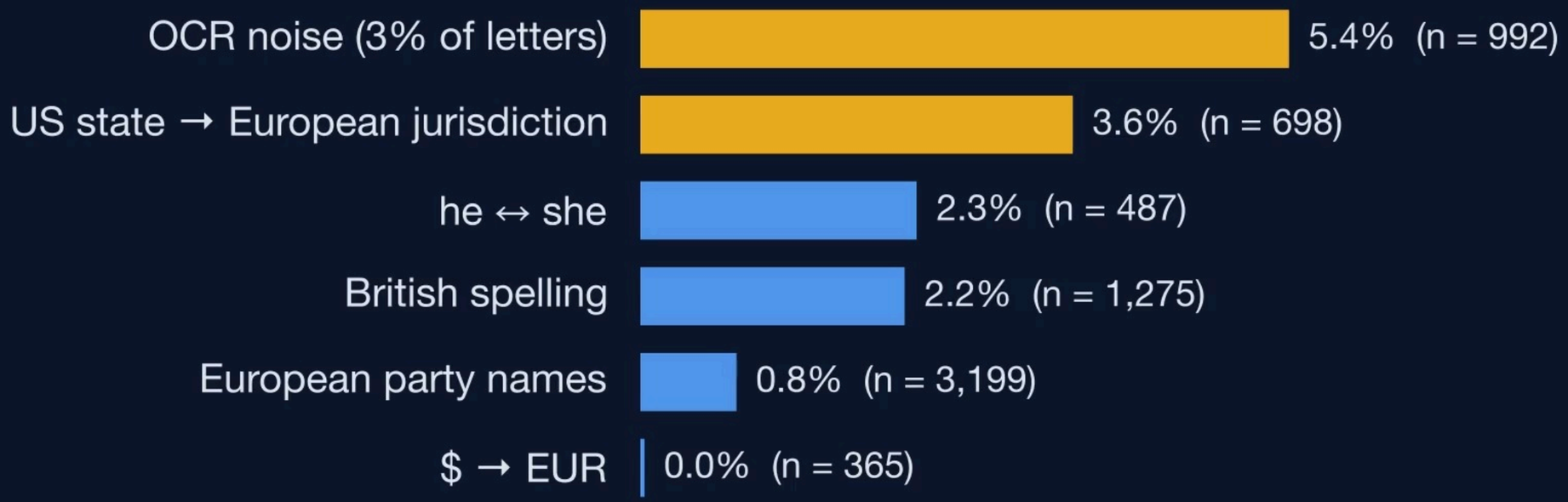


Abstention is not enough for leases and rare classes.

“she” (n = 37) and non-US (n = 36) groups: too small to conclude.

Audit: where the system is fragile (ensemble)

2 · Counterfactuals: does the predicted type change?



Type stability ≠ legal equivalence: changing the jurisdiction changes the applicable law.
e.g. Governing Laws → Waiver Of Jury Trials when “Delaware” becomes a European jurisdiction

Audit: where the system is fragile (ensemble)

3 · Out of scope: 8 inputs with no valid label

GDPR · force majeure · non-compete · French · German · …

Ensemble
abstains on 8 / 8

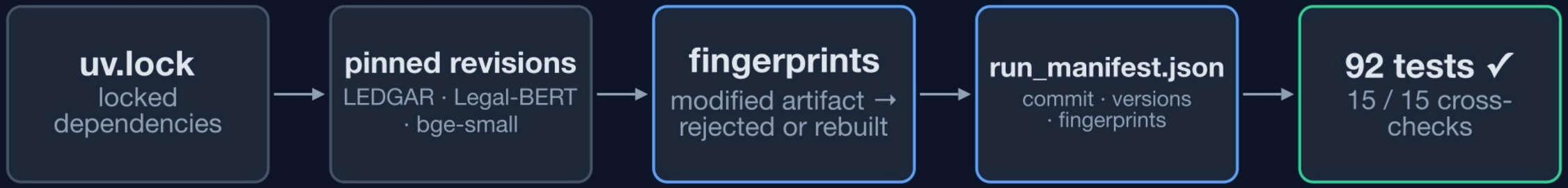
TF-IDF
abstains on 8 / 8

Legal-BERT alone
GDPR clause → “General”
confidence 0.90: confident error

A threshold reduces out-of-scope risk; it does not guarantee detecting it.

8 inputs: a small sample, not proof.

Reproducibility: every number is traceable



Demonstrated on this machine (Apple M3)

clean copy + uv sync --frozen → 92 tests passed

logits recomputed bit-identical

Not demonstrated

reproducibility on another machine

seed variance (a single seed)

Python 3.14 in beta

Permitted and prohibited uses

Intended use

- Help with **triaging** SEC-style **English** clauses
- **Systematic** human review
- Clauses below the threshold → lawyer
- Precedent search as a starting point

Do not use for

- Deciding a clause's type or scope alone
- European law, French, German: not validated
- Exhaustively detecting risks or unfair clauses
- Missing types: GDPR, force majeure, limitation of liability…
- Finding a specific notice period: non-semantic filter
- Drawing fairness conclusions about people: proxies

> Classifying a clause type is not judging its legal validity.

Limitations and next steps

Measured limitations

- BERT alone below TF-IDF; ensemble ≈ +0.7 pt
- Near-synonymous labels: *Applicable Laws / Governing Laws*, *Waivers / No Waivers*
- Lease clauses and rare classes
- Retrieval: the notion of notice period is not understood
- One seed, one machine, beta Python

Proposed next steps (not done)

- Controlled experiment: 256 vs 512 tokens, 2 vs more epochs with early stopping, several seeds
- Annotated European clauses, missing types
- Structured extraction of deadlines (role + duration + day type)
- Merging of near-synonyms validated by lawyers

Conclusion

88.43
ensemble test micro-F1 (macro 82.84)

84.2%
of clauses handled, 94.5% correct among them

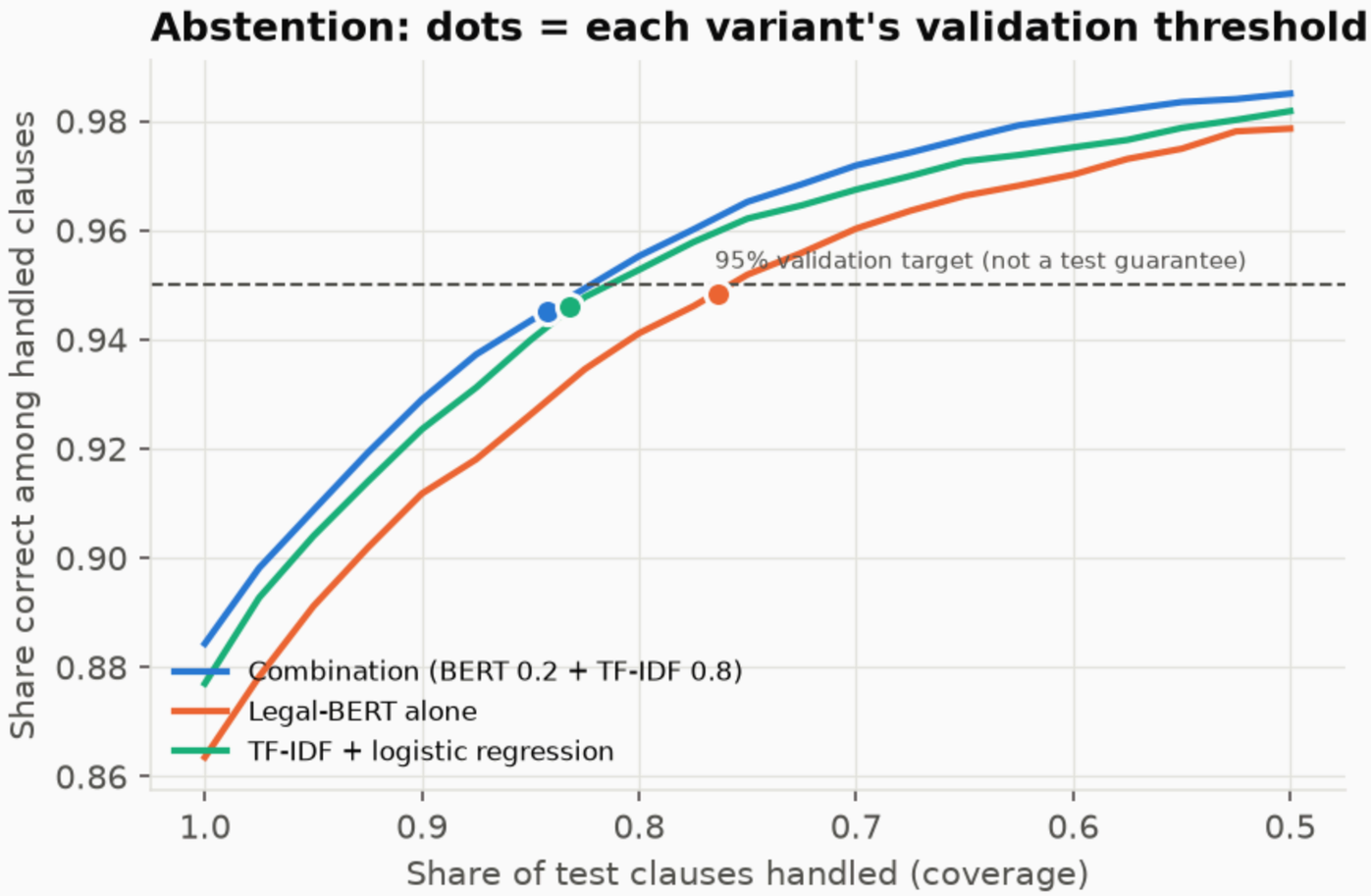
92
tests passed · traceable numbers

- A complete pipeline: data → models → validation → test → retrieval → audit
- An honest result: the ensemble beats a strong baseline **narrowly**
- A precise map of what the system cannot do

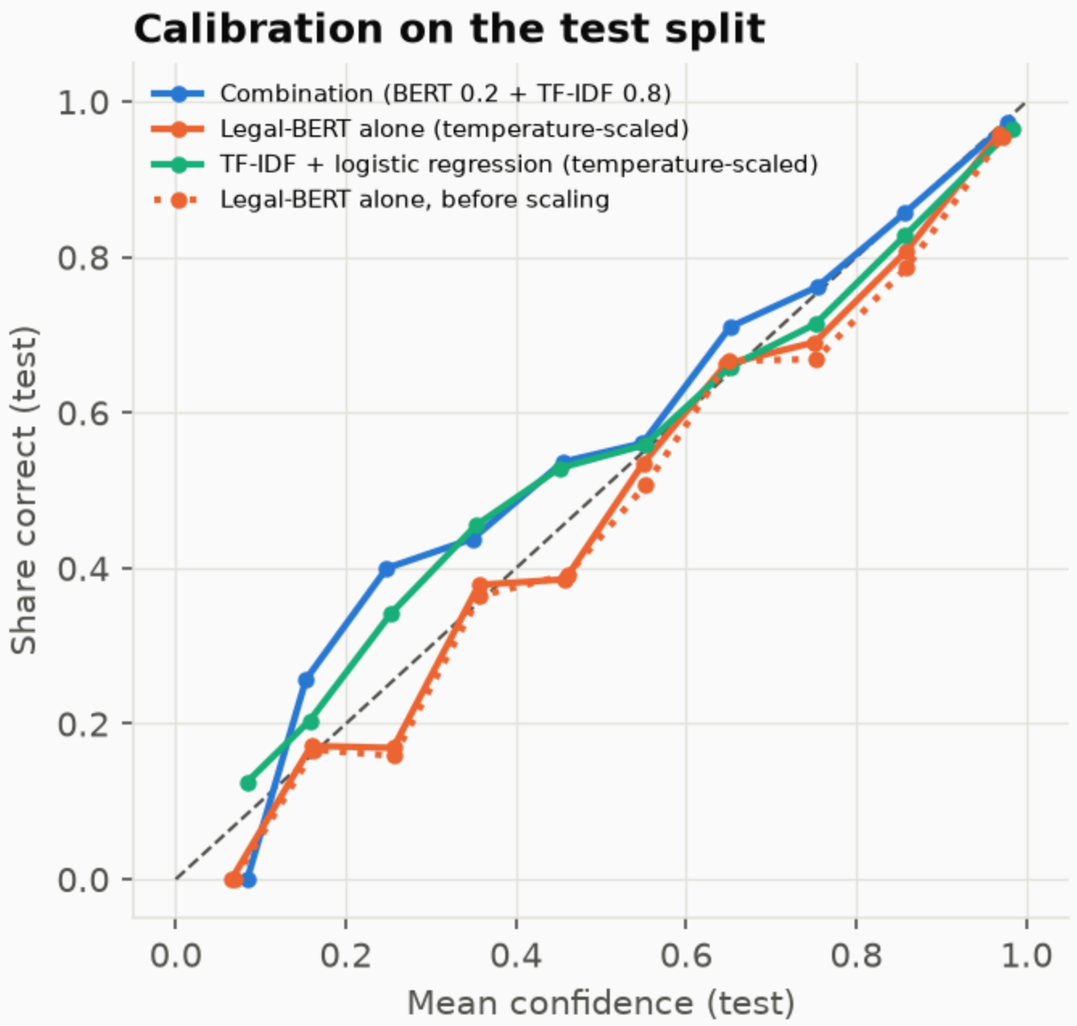
Recommendation: internal pilot, English clauses, lawyer in the loop, abstention enabled.

Useful for triage, not for decisions.

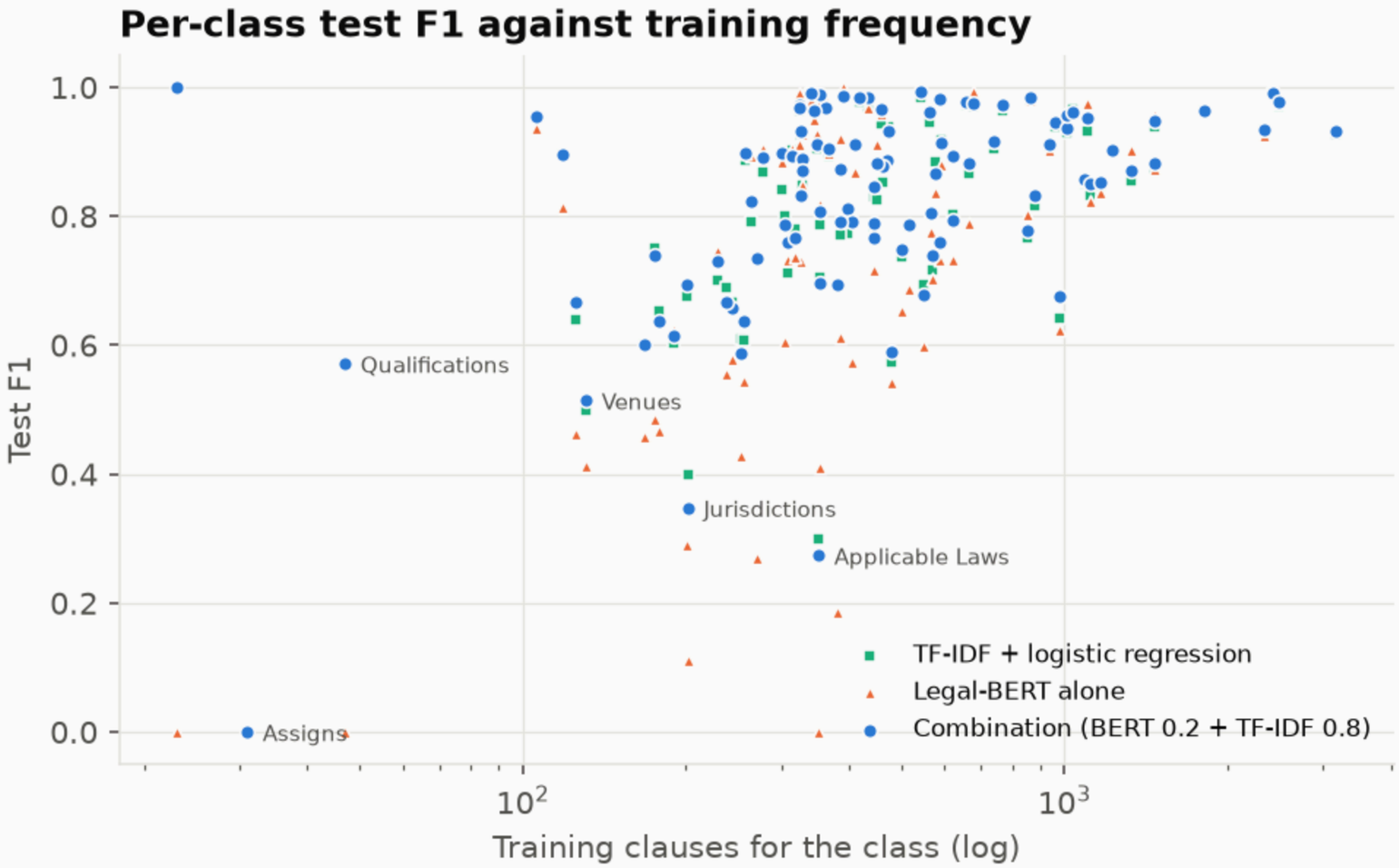
Coverage and accuracy among handled clauses (test)



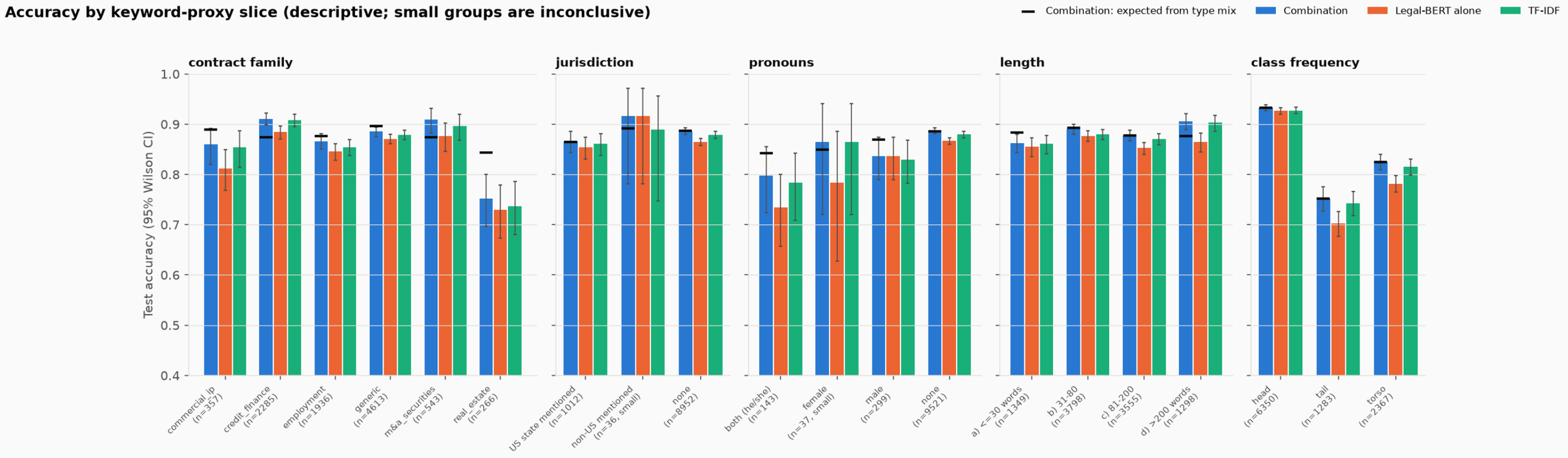
Calibration (reliability diagram)



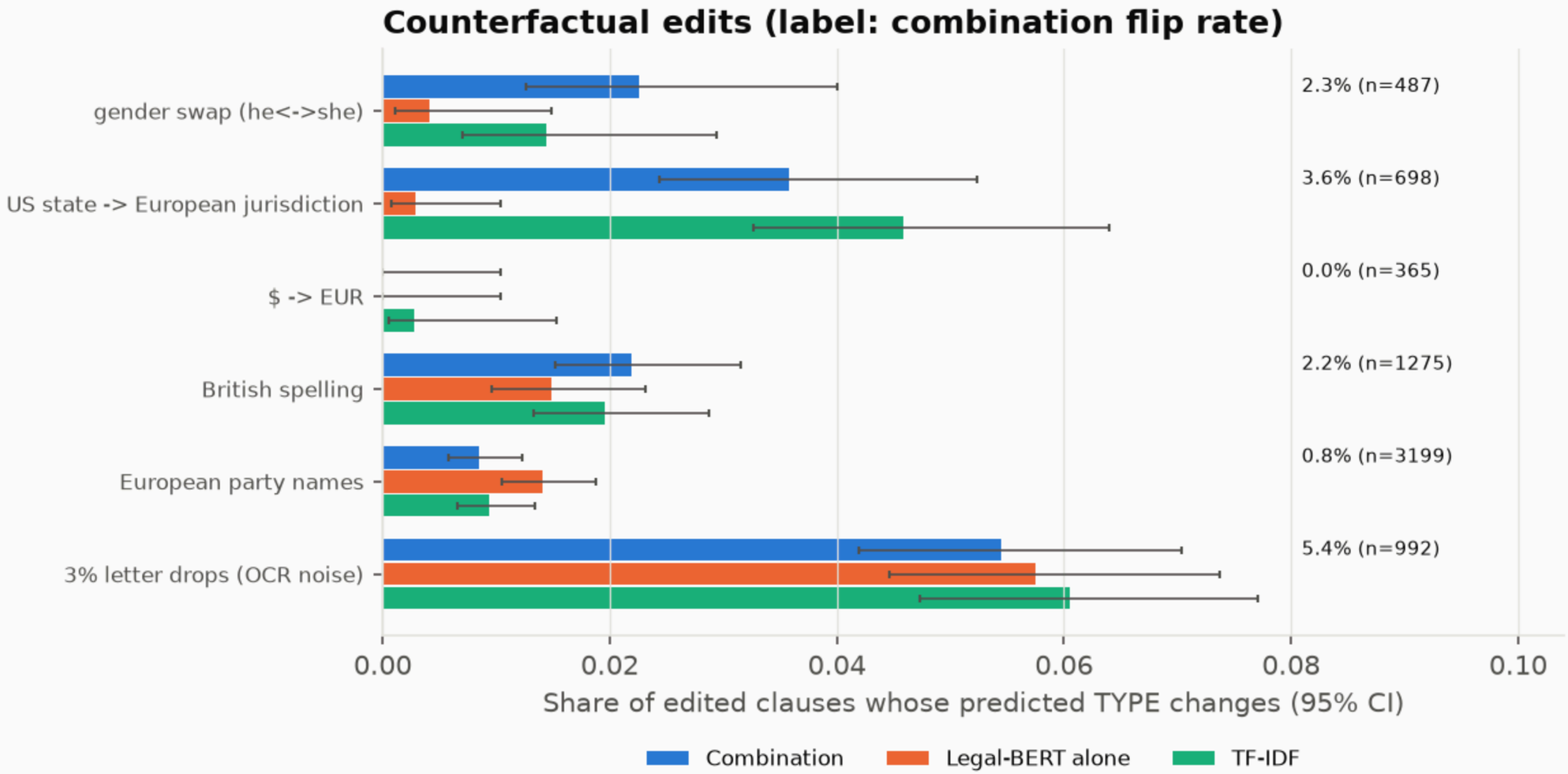
Per-class F1 by training frequency



Accuracy by group (keyword proxies)



Counterfactuals: predicted-type change rate



Data: class distribution and lengths

